

Atreya Mitra

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Open to Bengaluru / Hyderabad | Summer 2027 Internship

B.Tech CSE student at Manipal University Jaipur, Expected 2028. Builds and tests backend systems — authentication, distributed rate limiting, and payment verification.

EDUCATION

Manipal University Jaipur | Jaipur, Rajasthan

B.Tech in Computer Science and Engineering | Expected 2028 | CGPA: 8.16/10

Coursework/certifications: NPTEL Algorithms (IIT Madras, 2026), DSA in Java (IIT Kharagpur, 2025), SQL (Oracle Academy, 2025), Red Hat System Administration I & II (2026)

EXPERIENCE

Freelance Full-Stack Developer | Redline Garage | 2026

TanStack Start, TypeScript, PostgreSQL (Supabase), Razorpay

- Built a retailer-commissioned e-commerce platform with a PostgreSQL product/order schema covering catalog, cart, checkout, and order tracking.
 - Implemented server-side Razorpay HMAC signature verification with timing-safe comparison for payment integrity, plus inventory stock counts and low-stock indicators.
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PROJECTS

ScaleLink | Distributed URL Shortener and Rate Limiter

Node.js, Express.js, Redis, Lua, Nginx, Docker Compose, Azure, Terraform, Jest, k6

- Built a distributed sliding-window rate limiter using atomic Redis Lua scripts and server-side timing; verified 10 allowed / 190 rejected requests across two Node.js processes and correctness under 200 concurrent requests against real Redis.
- Deployed via Docker/Nginx multi-instance setup on Terraform-provisioned Azure VMs behind a load balancer with shared Redis; covered with 13 Jest tests, CI concurrency checks, and k6 load testing.

AuthForge | Authentication and Access Control Service

Node.js, Express.js, MongoDB, Redis, Docker, GitHub Actions, OpenAPI

- Built JWT authentication, bcrypt password hashing, RBAC, and TOTP two-factor auth with server-side signup-role enforcement and brute-force protection.
- Made refresh-token rotation atomic in MongoDB with session-version invalidation on logout; verified one successful rotation across 20 concurrent requests and correct rejection of stale tokens after logout.
- Covered refresh-token races and cookie-based logout with 26 Jest tests plus GitHub Actions integration checks against real MongoDB and Redis; documented REST APIs with OpenAPI/Swagger.

AccessLearn | Accessible Learning Platform

Next.js 14, TypeScript, Tailwind CSS, axe-core

- Built a full-stack platform that converts study material into accessible formats for users with different access needs, with a modular conversion pipeline and keyboard-accessible UI.
 - Audited with axe-core, fixed 2 detected violations (1 critical), and reached 0 accessibility violations across all application routes.
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LEADERSHIP

Operations Team, Oneiros 2025, Manipal University Jaipur — coordinated on-ground logistics and day-of scheduling for the university annual fest across multiple event days.

SKILLS

Languages & Frameworks: Java, Python, JavaScript, TypeScript, SQL, Node.js, Express.js, React, Next.js

Data: PyTorch, PostgreSQL, MongoDB, Redis, Supabase

Dev / SDLC: Docker, Docker Compose, Nginx, Azure, Terraform, Git, GitHub Actions, Jest, k6, Linux, CI/CD, OpenAPI/Swagger, unit & integration testing